Peter Mabrouk Morcos Mikhail

979-422-0640 | peter.morcos@tamu.edu | Urbana, IL | https://www.linkedin.com/in/petermab

Materials science researcher specializing in metal additive manufacturing, computational modeling, process optimization, and data-driven materials design, with a strong enthusiasm for leveraging AI to advance materials discovery and manufacturing innovation.

Education

Texas A&M University

- ***Ph.D. in Materials Science and Engineering***, School of Engineering, Texas A&M University, 2025.
  Thesis: "From Single Alloy to High Throughput Printability Assessment in Laser Powder Bed Fusion", **Advisor:** Dr Alaa Elwany.
  - "Data-Enabled Discovery and Design of Energy Materials (D3EM)" Certificate, Materials Science and Engineering Department, Texas A&M University.
  - "Applied Statistics – General" Certificate, Statistics Department, Texas A&M University.
  - "Academy for Future Faculty" Certificate, Center for Teaching Excellence, Texas A&M University.

- ***M.Sc. in Computational Mathematics (Non-Thesis)***, School of Science, Texas A&M University, 2024.
  Project: "Modeling Heat Transfer in Additive Manufacturing of Metals: A Finite Element Approach"

- ***M.Sc. in Materials Science and Engineering (Non-Thesis)***, School of Engineering, Texas A&M University, 2023.

The American University in Cairo

- ***M.Sc. in Nanotechnology***, School of Science and Engineering, The American University in Cairo (AUC), 2018.
  Thesis Topic: "Nanostructured Mg-ZK50 Sheets Fabricated for Potential Use for Biomedical Applications"
  Advisors: Dr Hanadi Salem, and Dr Khalil El-Khodary.

Suez Canal University (From Fall 2006 to Spring 2011)

- ***B.Sc. in Petroleum Exploration and Production Engineering,*** Faculty of Petroleum and Mining Engineering, Suez Canal University, 2011.

Professional Experience

University of Illinois Urbana-Champaign (UIUC) **(06/2025 – Present)**

- Post-doctoral Research Associate - Mechanical Science and Engineering Department
  Kinetic Lab - PI: Dr. Sameh Tawfick
  Conduct interdisciplinary research in metal additive manufacturing, computational modeling, process optimization, and advanced 3D printing technologies.
  Key Responsibilities and Research Projects
  1. Laboratory Development and Modeling Infrastructure:
  2. Mechanical Properties of Large-Format Metal Additive Manufacturing (Additive Friction Stir Deposition – AFSD)
  3. Effect of Process Parameters and Rheological Properties on Single-Line Deposition in Embedded 3D Printing *(In progress)*
  4. Image Segmentation for Melt Pool Characterization in Laser Powder Bed Fusion (LPBF) *(In progress)*
  5. Automated Calibration Tool for AdditiveFOAM *(In progress)*

Texas A&M University **(08/2019 – 05/2025)**

- Graduate Teaching Assistant (GAT) - Materials Science and Engineering Department
  Courses:

MSEN 400 Design and Analysis of Materials Experiments
MSEN 330 Numerical Methods for Materials Scientists and Engineers

- Graduate Research Assistant - Materials Science and Engineering Department
  Additive Manufacturing at Texas A&M (AMTEX) Group, PI: Dr. Alaa Elwany (Professor at the Department of Industrial & Systems Engineering)
  Key Responsibilities and Research Projects
  1. Laboratory Operations and Technical Support
  2. Efficient Alloy and Process Design for Additive Manufacturable Refractory Alloys*;* PI: Dr. Raymundo Arroyave, Co-PI: Dr. Alaa Elwany, Dr. Ibrahim Karaman (Texas A&M University), and Dr James Mabe (The Boeing Company).
  3. Additive manufacturing powder research for Alloy 718 (IN718); PI: Dr. Alaa Elwany, Co-PIs: Dr. Raymundo Arroyave, Dr. Ibrahim Karaman (Texas A&M University).
- Graduate Teaching Assistant - Engineering Technology and Industrial Distribution Department
  Courses:
  MMET 376: Strength of Materials Lab

The American University in Cairo **(09/2011 – 05/2019)**

- Teaching Assistant - Petroleum & Energy Engineering Department
  Courses:
  Drilling Simulation Lab
  Drilling Engineering I
  Drilling Engineering II
  Advanced Well Construction
  Reservoir Rock and Fluid Properties Lab
  Reservoir Engineering and Recovery
  Reservoir Economics, Management and Risk Analysis

Gulf of Suez Petroleum Company **(07/2012 – 08/2019)**

Positions:

- Drilling Performance Engineer (1 year)
- High Risk Drilling Engineer (1 year)
- Well Planning and Design Engineer (1.5 years)
- Offshore Well Site Drilling Engineer (2 years)
- Operations Drilling Engineer (0.5 years)
- Drilling Contracts Engineer (1 year)

Technical Skills

Experimental Skills

1. Operating additive manufacturing machines (L-PBF, LW-DED)
2. Metal sample preparation
3. Optical microscopy
4. Scanning Electron microscopy
5. Energy dispersive X-ray spectroscopy
6. X-ray diffraction
7. Mechanical testing

Modelling and Simulation Skills

1. Python (Advanced level, used in modelling of AM process and machine learning applications)
2. OpenFOAM (Intermediate level, used in AM modelling (L-PBF, microstructure, and embedded 3d printing)
3. C++ (Beginner level, used in modeling heat equation using deal.II library)
4. Flow-3D (modelling single tracks, multi-tracks, multi-layers, and multi-materials for LPBF and LW-DED processes)
5. MatLab (beginner level, used in coding different linear algebra methods)

6. Solidworks (beginner level, used in CAD drawings)
7. Ansys (Intermediate level, used in AM modelling)
8. PravaH AM (modelling single tracks, multi-tracks, and multi-materials for LPBF and LW-DED processes, and microstructure modelling)

Awards and Grants:

AUC Honor Certificate	**May 2016:** Received AUC Honor Certificate for academic achievements. **May 2017:** Received AUC Honor Certificate for academic achievements.
AUC Conference Grant	**December 2016:** Received AUC conference grant to travel to California, USA for presenting my paper titled "Mechanically Alloyed Magnesium Based Nanostructured Alloy Powders for Biomedical Applications" in TMS conference.
AUC Research Grant	**May 2015:** Received AUC research grant for the topic "Studying Magnesium Alloys with nano-sized grains for application in Cardiac Stents"
AUC Graduate Fellowship	**January 2012:** Received Laboratory Instruction Graduate Fellowship from AUC for studying M.Sc. degree in Nanotechnology at AUC.
Egypt's Nobel Prize Simulation Award	**July 2011:** Participated in Nobel Prize Simulation 3rd Annual Conference with a research paper "Application of Casing while Drilling in Drilling Offshore Wells in Nile Delta, Egypt" and awarded the ***first place*** in the Engineering Section.

Publications

Published Papers

1. S. Sheikh; B. Vela; P. Honarmando; **P. Morcos**; D. Shoukr; A. M. Kotb; I. Karaman; A. Elwany, R. Arroyave; "High-throughput alloy and process design for metal additive manufacturing"; npj Computational Materials; volume 11; (2025); https://doi.org/10.1038/s41524-025-01670-x
2. S. Sheikh; B. Vela; P. Honarmando; **P. Morcos**; D. Shoukr; I. Karaman; A. Elwany, R. Arroyave; "An automated computational framework to construct printability maps for additively manufactured metal alloys"; npj Computational Materials, volume 10; (2024); https://doi.org/10.1038/s41524-024-01436-x
3. **P. Morcos**; B. Vela; C. Acemi; I. Karaman; A. Elwany, R. Arroyave; "Data-augmented modeling in laser powder bed fusion: A Bayesian approach", Additive Manufacturing; volume 96, (2024); https://doi.org/10.1016/j.addma.2024.104545
4. S. Sheikh; B. Vela; V. Attari; X. Huang; **P. Morcos**; J. Hanagen, C. Acemi; I. Karaman; A. Elwany, R. Arroyave; "Exploring chemistry and additive manufacturing design spaces: a perspective on computationally-guided design of printable alloys", Materials Research Letters; volume 12, (2024); https://doi.org/10.1080/21663831.2024.2316204
5. **P. Morcos**; D. Shoukr; T. Sundermann; T. Dobrowolski; C. Yates; N. Barta; J. R Jain, R. Arróyave, I. Karaman; A. Elwany; "An all-encompassing study on the joint effect of powder feedstock characteristics and manufacturing process parameters on the densification and mechanical properties of additively manufactured nickel alloy 718"; Additive Manufacturing; volume 79, (2023); https://doi.org/10.1016/j.addma.2023.103828
6. D. Shoukr; **P. Morcos**; T. Sundermann; T. Dobrowolski; C. Yates; J. R Jain; R. Arróyave; I. Karaman, A. Elwany; "Influence of layer thickness on the printability of nickel alloy 718: A systematic process optimization framework", Additive Manufacturing, Volume 73, (2023); https://doi.org/10.1016/j.addma.2023.103646
7. **P. Morcos;** A. Elwany; I. Karaman; R. Arroyave; "Review: Additive Manufacturing of Tungsten and Tungsten-based Alloys", Journal of Materials Science, Volume 57, (2022); https://doi.org/10.1007/s10853-022-07183-y
8. P. Honarmandi; R. Seede; L. Xue; D. Shoukr; **P. Morcos;** B. Zhang; C. Zhang; A. Elwany; I. Karaman; R. Arroyave; "A rigorous test and improvement of the Eagar-Tsai model for melt pool characteristics in laser powder bed fusion additive manufacturing", Additive Manufacturing, Volume 47, (2021); doi:10.1016/j.addma.2021.102300

9. **Morcos, P. M.**; El-Khodary K. I.; and Salem, H. G.; "Mechanically Alloyed Magnesium Based Nanostructured Alloy Powders for Biomedical Applications", Magnesium Technology. Proc. of The Minerals, Metals and Materials Society 146th Annual Meeting and Exhibition (TMS 2017), USA, San Diego, California; (2017); doi: 10.1007/978-3-319-52392-7_9.
10. **Morcos, P.M.**; "Casing while Drilling and its Implementation in Drilling Offshore Wells in Egypt"; 10th **O**ffshore **M**editerranean **C**onference (OMC) Proceeding, March (2011), paper number **DRILL/EO/03.**

Conference Presentations

1. **P. Morcos;** B. Vela; C. Acemi; A. Elwany; I. Karaman; R. Arroyave; "Towards High-Throughput Assessment of Printability in Refractory Alloys Systems for Laser-Powder Bed Fusion", The 34th Annual International Solid Freeform Fabrication Symposium (SFF 2023), Austin, Texas, USA.
2. P. Morcos; M. Vaughan; "Predicting the Printable Parameter Space for Laser Directed Energy Deposition Using a Calibrated Model" The 34th Annual International Solid Freeform Fabrication Symposium (SFF 2023), Austin, Texas, USA.
3. B. Vela; P. Morcos; C. Acemi; I. Karaman; A. Elwany; R. Arroyave; "Bayesian Data-augmentation of Thermal Models for Design of Nb-Ta-W Alloys", The 34th Annual International Solid Freeform Fabrication Symposium (SFF 2023), Austin, Texas, USA.
4. S. Sheikh; B. Vela; P. Honarmandi; **P. Morcos;** D. Shoukr; A. Kotb; I. Karaman; A. Elwany; R. Arroyave; "High-Throughput Computation and Process Design for Metal Additive Manufacturing", 7th World Congress on Integrated Computational Materials Engineering (ICME 2023), Brisbane, Australia.
5. **P. Morcos;** B. Vela; C. Acemi; A. Elwany; I. Karaman; R. Arroyave; "Towards High-Throughput Assessment of Printability in Refractory Alloys Systems for Laser-Powder Bed Fusion", The Minerals, Metals & Materials Society Conference (TMS 2023), San Diego, California, USA.
6. C. Acemi; W. Trehern; E. Norris; B. Vela; **P. Morcos;** R. Arroyave; A. Elwany; I. Karaman; "High-throughput design, synthesis, and characterization of W-based refractory multi-principal element alloys (MPEAs)", The Minerals, Metals & Materials Society Conference (TMS 2023), San Diego, California, USA.
7. S. Sheikh; P. Honarmandi; B. Vela; **P Morcos;** D. Shoukr; R. Arroyave; A. Elwany; I. Karaman; "A Calibration-Free Physics-based Framework to Predict Printability Maps in Additive Manufacturing Process", The Minerals, Metals & Materials Society Conference (TMS 2023), San Diego, California, USA.
8. **P. Morcos;** D. Liu; A. Elwany; I. Karaman; R. Arroyave; "Bayesian Process Optimization for Porosity Control in Laser-Powder Bed Fusion of IN718 Alloy with Computational Fluid Dynamics Simulation", The Minerals, Metals & Materials Society Conference (TMS 2023), San Diego, California, USA.
9. **P. Morcos;** D. Shoukr; T. Sundermann; T. Dobrowolski; N. Barta; J R. Jain; R. Arroyave; I. Karaman; A. Elwany; "An All-encompassing Study on the Joint Effect of Powder Feedstock Characteristics and Manufacturing Process Parameters on the Densification and Mechanical Properties of Additively Manufactured Nickel Alloy 718", The Minerals, Metals & Materials Society Conference (TMS 2023), San Diego, California, USA.
10. T. Sundermann; D. Shoukr; **P. Morcos;** R. Arroyave; A. Elwany; I. Karaman; "Printability and Failure Susceptibility of Different Powder Layer Thicknesses in Laser Powder Bed Fusion", The Minerals, Metals & Materials Society Conference (TMS 2023), San Diego, California, USA.
11. D. Shoukr; **P. Morcos;** R. Arroyave; A. Elwany; "Material-agnostic Spatiotemporal Modeling of the Thermal Field History in the L-PBF Additive Manufacturing Process", 2022 INFORMS Annual Meeting, Indianapolis, Indiana, USA.
12. S. Sheikh; B. Vela; **P Morcos;** P. Honarmandi; R. Arroyave; A. Elwany; I. Karaman; "Designing Calibration-Free Physics-Based Printability Maps for Additively Manufactured Cantor Alloys", The Annual International Solid Freeform Fabrication Symposium (SFF 2022), Austin, Texas, USA.
13. S. Sheikh; P. Honarmandi; B. Vela; **P Morcos;** D. Shoukr; R. Arroyave; A. Elwany; I. Karaman; "A Calibration-Free Physics-based Framework to Predict Printability Maps in Additive Manufacturing Process" The Annual International Solid Freeform Fabrication Symposium (SFF 2022), Austin, Texas, USA.
14. D. Shoukr; **P. Morcos;** I. Karaman; R. Arroyave; A. Elwany; "Spatiotemporal Modeling of the L-PBF Thermal Field History" The Annual International Solid Freeform Fabrication Symposium (SFF 2022), Austin, Texas, USA.
15. **P. Morcos;** D. Shoukr; T. Sundermann; T. Dobrowolski; N. Barta; J R. Jain; R. Arroyave; I. Karaman; A. Elwany; "A Study on the Effect of VED, Particle Size Distribution, Moisture content, and Powder Reuse on the

Densification and Mechanical Properties of L-PBFed Nickel Alloy 718 Using Design of Experiment and ANOVA", The Annual International Solid Freeform Fabrication Symposium (SFF 2022), Austin, Texas, USA.

16. **P. Morcos;** D. Shoukr; T. Sundermann; T. Dobrowolski; C. Yates; J R. Jain; R. Arroyave; I. Karaman; A. Elwany; "Printability of Nickel Alloy 718 Using a Systematic Process Optimization Framework with Different Layer Thicknesses", The Annual International Solid Freeform Fabrication Symposium (SFF 2022), Austin, Texas, USA.
17. **Morcos, P. M.;** Hassanein, N. A.; Amleha, A.; and Salem, H. G.; "Manufacturing of Nanostructured Mg-Zn-Zr Wrought Sheets for Potential Application as a Biomaterial", Materials Science and Technology Conference (MS&T 2018), USA, Columbus, Ohio.
18. **Morcos, P.M.**; "Application of casing while Drilling in Drilling Offshore Wells in Nile Delta, Egypt", 2012 Young Professionals Technical Symposium, SPE-Saudi Arabia Section (SPE-SAS YPTS) March 2012, Dhahran, Saudi Arabia. Egypt" Poster Presentation in 2012 AUC Research Conference: Entrepreneurship and Innovation,
19. **Morcos, P.M.**; "Application of Casing while Drilling in Nile Delta, Shaping the Future of Egypt", 2012 AUC Research Conference: Entrepreneurship and Innovation, Shaping the Future of Egypt, AUC, Cairo, Egypt.
20. **Morcos, P.M.**; "Casing while Drilling and its Implementation in Drilling Offshore Wells in Egypt", 10th **O**ffshore **M**editerranean **C**onference (OMC), March 2011, paper number **DRILL/EO/03.,** Ravenna, Italy.

Conference Posters

1. **P. Morcos**; A. Rashid; W. King; S. Tawfick; "Towards a Universal Scaling Law for Predicting Contact Angle in Laser Powder Bed Fusion" The Minerals, Metals & Materials Society Conference (TMS 2026), San Diego, California, USA.
2. A. Rashid; **P. Morcos**; S. Ozerinc; S. Tawfick; W. King; "Fatigue and Tensile Properties of Ti-6Al-4V Fabricated with Wire-feed Electron Beam Additive Manufacturing" The Minerals, Metals & Materials Society Conference (TMS 2025), San Diego, California, USA.
3. **P. Morcos;** D. Shoukr; T. Sundermann; T. Dobrowolski; C. Yates; J R. Jain; R. Arroyave; I. Karaman; A. Elwany; "Printability of Nickel Alloy 718 Using a Systematic Process Optimization Framework with Different Layer Thicknesses", The Minerals, Metals & Materials Society Conference (TMS 2023), San Diego, California, USA.
4. M. Vaughan; M. Elverud; J. Ye; **P. Marcos**; R. Seede; S. Gibbons; R. Arroyave; A. Elwany; I. Karaman; "The Development of a Directed Energy Deposition (DED) Printability Framework for Improving Part Density and Performance in a 316L Stainless Steel", The Minerals, Metals & Materials Society Conference (TMS 2022), Anaheim, California, USA.

Summer Schools and Courses

Summer Schools:

1. ***June 27th to July 1st, 2022*:** SATA 2022 - School for Advanced Thermodynamic Assessments, CIRIMAT Laboratory, University of Toulouse, Toulouse, France**.**
2. ***June 29th to July 10th, 2020:*** CMS3 2020 - Computational Materials Science Across Scales, Texas A&M University, College Station, Texas, USA.
3. ***29 August to 2 September, 2016:*** PISACMS 2016 – Paris International School on Advance Computational Material Science, University Pierre and Marie Curie, Paris, France.
4. ***2 June to 11 June, 2016:*** PIRE-CNA 2016 Summer School: New Frontiers in Nonlinear Analysis for Materials, Carnegie Mellon University (CMU), Pittsburg, Pennsylvania, USA.

Service Outreach

- Volunteered in "ISUR Research Expo" as a judge in poster presentations session, University of Illinois Urbana-Champaign, Spring 2026
- Volunteered in "Student Research Week" as a judge in poster presentations session, Texas A&M University, Spring 2022
- Invited speaker at AUC in Well Completion courses to transfer field experience to students
- Invited Speaker at American Association for Petroleum Geologists, Suez Canal University student chapter.
- Invited Speaker in SPE SCU SC for undergraduate students in drilling and nanotechnology topics
- August 2011, Scientific Presenter in Science Lantern Event, AUC
- SPE North Africa Paper Contest Manager, Season 2010/2011

- SPE Cairo University Local Paper Contest Manager, Season 2010/2011
- Official Media Representative in Egyptian Inventors Union
- Delegate in Model Council of Ministers in the Ministry of Education, AUC, Season 2011
- Participated in the National Conference and Exhibition for Creativity and Technology Development with a research paper entitled "Casing While Drilling – An Overview", October 2010
- Participated in the 1st Cairo Science Festival at AUC, with a presentation titled "Casing While Drilling: An Overview"
- Participated in the 1st Business Technical Platform organized by SPE CU SC, March, 2010
- Active Member in SPE CU SC Season 2009/2011, Academic Committee

References

1. Dr Alaa Elwany; "Professor at Texas A&M University, Department of Industrial & Systems Engineering", PhD advisor.
   E-mail: elwany@tamu.edu
2. Dr. Ibrahim Karaman; "Professor at Texas A&M University, Department of Materials Science and Engineering", PhD committee member.
   E-mail: ikaraman@tamu.edu
3. Dr. Sameh Tawfick; "Professor at University of Illinois Urbana-Champaign, Department of Mechanical Science & Engineering", Postdoctoral PI.
   E-mail: tawfick@illinois.edu
4. Dr. Bill King; "Professor at University of Illinois Urbana-Champaign, Department of Mechanical Science & Engineering", Postdoctoral Co-PI.
   E-mail: wpk@illinois.edu